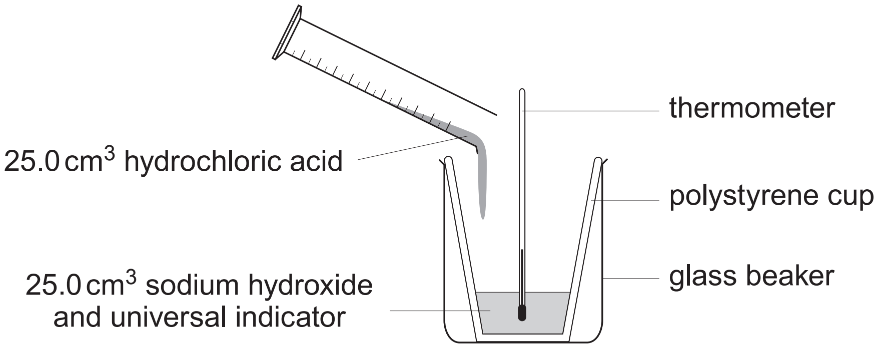
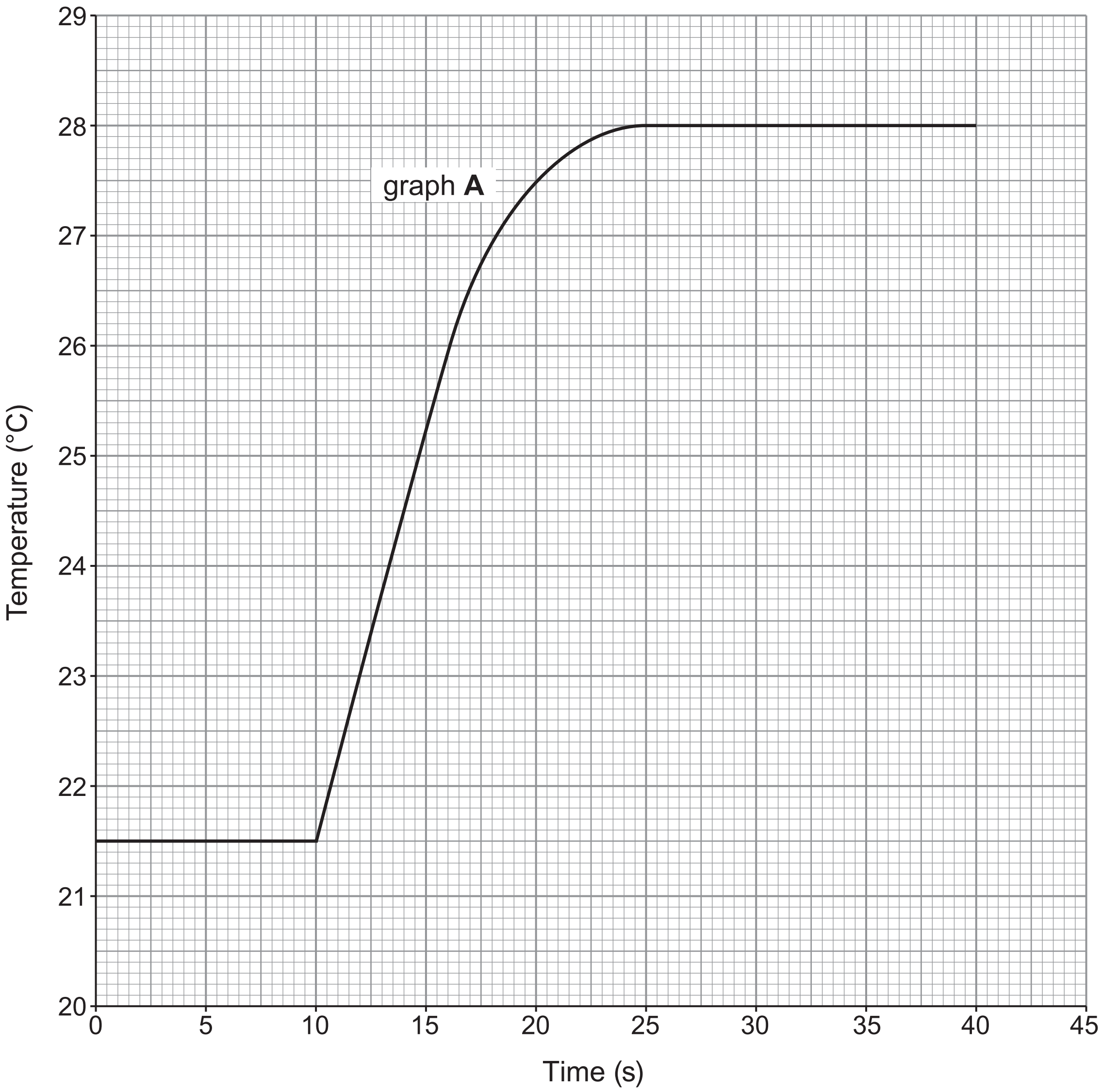
8. A student investigated the temperature rise during a neutralisation reaction.



The student put 25.0 cm^3 of sodium hydroxide solution and 5 drops of universal indicator into a polystyrene cup and recorded the temperature of the alkali. After 10 seconds the student added 25.0 cm^3 of dilute hydrochloric acid to the alkali and recorded the temperature every 5 seconds for another 30 seconds. Graph **A** shows the results obtained.



Examiner only

(a) (i) Use the graph to find the maximum temperature rise during the reaction. [1]

Temperature rise = ... °C

(ii) The energy given out can be calculated using the following formula.

energy given out = **total** volume of reaction mixture $\times$ 4.2 $\times$ temperature rise

Calculate the energy given out during the reaction. [2]

Energy given out = ... J

(iii) The temperature of the contents in the cup was recorded after 2 hours. Give the final temperature reading you would expect. Give the reason for your answer. [1]

Final temperature .... °C

Reason ....

....

(b) The student repeated the experiment using 25.0 cm^3 of ethanoic acid of the same concentration as the hydrochloric acid. The table shows the results obtained.

Time (s)	0	5	10	15	20	25	30	35	40
Temperature (°C)	21.5	21.5	21.5	24.0	26.0	26.9	27.0	27.0	27.0

Plot the results on the grid on page 26. Draw a suitable line. Label your graph **B**. [2]

(c) Use the graphs to state which of the two acids is the stronger – hydrochloric acid or ethanoic acid. Give the reason for your choice. [1]

Acid ....

Reason ....

....

Examiner only

(d) The temperature rises in both experiments were much **lower** than expected. The student suggested that using a temperature sensor instead of a thermometer would give temperature rises closer to the expected values.

(i) State why using a temperature sensor would still give a lower than expected temperature rise. [1]

....

....

(ii) What improvement to the apparatus would you suggest to the student to obtain temperature rises closer to the expected values? [1]

....

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